
Nearby Evidence Shapes How Transformers Respond to Planted Errors in Their Own Reasoning

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Abstract

1 Can an autoregressive language model recover from a planted error in
2 its own reasoning without external feedback? Simple compounding-error
3 arguments often assume that false intermediate steps are unrecoverable.
4 We test this assumption from the inside by intercepting a proof the model
5 already solves, planting a single false step, and formally validating the sub-
6 sequent reasoning trace. We find that recovery is strongly shaped by local
7 verification cost. When a false step contradicts nearby evidence, models fre-
8 quently catch and correct the error, often voicing explicit doubt. Conversely,
9 errors requiring broader proof search are often silently absorbed and built
10 upon. This falsifiability gradient holds across models from 1.5B to 32B pa-
11 rameters. While targeted control experiments support a causal role for local
12 evidence in explicit error-flagging and reduced injection-dependence, they
13 also reveal the difficulty of fully isolating these dynamics from grammatical
14 and lexical confounds. Ultimately, our results refine the compounding-
15 error debate: per-step errors do compound silently, but primarily when
16 they evade local checkability.

17 1 Introduction

18 Autoregressive language models generate one token at a time, so long-horizon reasoning
19 can be fragile. A standard compounding argument says that if each step is correct with
20 probability $(1 - \epsilon)$ and a wrong step can never be repaired, then the chance that a T -step
21 derivation is fully correct decays like $(1 - \epsilon)^T$ (LeCun, 2022; Dziri et al., 2023). The argument
22 relies on independence, a constant error rate, and unrecoverability; the last premise carries
23 much of the force. Rather than add an external verifier, tool, or search procedure, we ask
24 what happens inside a single generation. If we plant one error in a model’s own proof and
25 remove external feedback, does the model build on the error, route around it, or complete a
26 valid proof anyway?

27 Prior work gives reasons to expect both fragility and robustness. Interventional studies
28 perturb visible chain-of-thought and report surprising robustness (von Recum et al., 2026);
29 mechanistic analyses find redundant, self-repairing structure inside networks (McGrath et
30 al., 2023; Lad et al., 2024). Neither line fully settles our question, because two details matter
31 here: whether the planted step is actually false, and whether apparent recovery is a genuine
32 proof or a restatement of a prompted goal. Once those are controlled, the central pattern
33 is simple: the model is more likely to avoid an error when the surrounding proof contains
34 nearby evidence against it, and more likely to use the planted fact when detecting the error
35 requires a broader proof search. In the expanded Qwen2.5-7B sweep, globally checkable
36 falsehoods yield closure-valid completion rates of 0.360–0.447 across injection positions and
37 injection-dependence rates of 0.233–0.420, while one-hop falsehoods remain 0.633–0.740
38 closure-valid and produce much higher verbalized doubt.

39 Contributions.

- 40 1. **Perturb-and-trace protocol.** We plant a controlled error in a model’s own proof and
41 use a formal validator to classify closure-valid completion, injection-dependence,

42 goal restatement, or derailment. This distinction matters because final-answer
43 accuracy can count parroted goals as recovery.

44 2. **Truth-status audit.** We audit every planted step, preventing off-path-but-entailed
45 facts from being counted as errors.

46 3. **Nearby-evidence conditions.** We compare planted steps that range from meaning-
47 preserving paraphrases to errors contradicted by nearby statements and errors
48 detectable only from the full rule closure. All rates and confidence intervals are
49 generated from archived result artifacts.

50 4. **Controls and limitations.** Certificate and polarity-control experiments support
51 a role for local evidence but also expose boundary conditions: local certificates
52 increase explicit doubt and reduce injection-dependence, while the strict lexical-
53 polarity design fails to produce a clean local false positive categorical cell in this
54 grammar.

55 2 Setup

56 **Evaluation domain.** We use PrOntoQA-OOD (Saparov and He, 2023), which generates
57 fictional ontologies (e.g., “every wumpus is a tumpus”) and asks the model to derive a
58 target proof. Its rigid grammar lets a proof validator parse each sentence. We use a fixed
59 PrOntoQA-OOD cohort with multi-hop proofs.

60 **Model cohort.** We evaluate six open-weights models to isolate scale, architectural lineage,
61 and reasoning-specific training: Qwen2.5-Instruct (1.5B, 7B, 32B), OLMo-2-7B-Instruct,
62 Llama-3.1-8B-Instruct, and DeepSeek-R1-Distill-Qwen-7B. All models are run greedily and
63 without fine-tuning, with Qwen2.5-7B serving as our primary subject. We perturb only
64 proofs that the model successfully solves and that pass the grammar validator. Because
65 every injected trajectory starts from a successful proof, the reported closure-valid completion
66 rates are upper bounds.

67 **The perturb-and-trace protocol.** The model first writes a correct proof. We then intercept
68 the proof, select one intermediate step (early, middle, or late), and overwrite it with a
69 planted sentence. The model resumes generation with the same token limits, no external
70 feedback, and no prompt asking it to self-correct. We record the full continuation and
71 validate what it writes next. For latent comparisons, we also sample eight temperature-0.8
72 continuations from the same interception point with the correct step left intact, retaining
73 only valid continuations. Hidden states are cached at five depths throughout the sequence.

74 **Planted-step conditions.** The independent variable is the local verification cost of the
75 planted step. We use “falsifiability gradient” to mean the distance, in local proof context and
76 rule closure, between a planted claim and the nearest evidence that refutes it. We structure
77 six perturbations into this gradient, ordered by how easily the step can be checked against
78 the surrounding proof:

- 79 (a) **Benign paraphrase:** The correct step, reworded. Controls for the baseline cost of
80 interrupting the generation trace.
- 81 (b) **Off-path category (True interruption):** A fact absent from the gold proof path but
82 logically entailed by the world premises.
- 83 (c) **Distractor:** A fabricated rule entirely irrelevant to the active entity.
- 84 (d) **Contradiction:** The negation of a fact already stated in the proof. This is false and
85 conflicts with adjacent context at distance zero.
- 86 (e) **Negated step:** The negation of the step about to be derived. This is false and
87 conflicts with no prior statement, but is refutable via a single-hop inference from
88 the surrounding proof.
- 89 (f) **Globally-checkable falsehood:** A categorical claim that is strictly unentailed. It
90 lacks nearby conflicting evidence and is detectable only by searching the entire rule
91 closure.

2.1 Auditing Perturbation Validity

In entailment-dense PrOntoQA worlds, a fact absent from the gold proof can still be true. We therefore audit every planted atom against the rule closure and report only families with verified truth status. Because globally-checkable falsehoods are rarer under the constrained generator, the primary sweep expands that family to 150 examples per injection position and matches it with true-interruption, benign-paraphrase, and one-hop-falsehood controls (Table 1).

Condition	n	Closure-valid [95% CI]	Inj.-dep. [95% CI]	Doubt [95% CI]	Unparsed
true interruption	450	0.856 [0.818,0.889]	0.000 [0.000,0.000]	0.020 [0.007,0.036]	0.013
benign paraphrase	450	0.849 [0.813,0.882]	0.000 [0.000,0.000]	0.080 [0.056,0.107]	0.024
1-hop falsehood	450	0.671 [0.624,0.722]	0.016 [0.004,0.029]	0.336 [0.291,0.387]	0.038
global falsehood	450	0.393 [0.342,0.442]	0.342 [0.293,0.384]	0.013 [0.004,0.024]	0.018

Table 1: Expanded audited primary PrOntoQA perturbation set. Each condition pools three injection positions (150 examples per position); confidence intervals are problem-cluster bootstrap intervals.

2.2 Metrics for Completion and Failure

Because a model can print the correct final line via heuristic shortcuts while producing an invalid proof, final-answer accuracy can overstate recovery. We therefore formally parse and validate the proof trace itself.

A continuation is classified as a **closure-valid completion** if every stated fact follows from the true premises and prior valid steps under the rule closure, culminating in the target. This metric measures semantic validity under closure, not faithful replay of the original derivation. We categorize failure modes as follows:

Injection-dependent: Subsequent reasoning follows only if the planted atom is assumed as true. For genuinely false, positive-category injections, this constitutes strict poisoning.

Parroted: The target is reached, but the proof contains an untailed intermediate step.

Derailed: The continuation ends on an incorrect statement.

To quantify explicit error-flagging, we scan for hedging and self-correction tokens (e.g., “wait,” “however,” “contradict”). This heuristic detector serves strictly as a behavioral flag for doubt and is not treated as proof of logical repair.

Validator Limitations: The proof validator is closure-based: it checks whether each fact follows from the true premises and prior valid steps, but it allows the model to skip steps. We found that closure-valid continuations can include derivation-free jumps to the goal; at late positions, the proof has fewer intermediate steps left to skip. Furthermore, strict poisoning is only mechanically detectable for positive category claims, as the grammar derives nothing from a negation. Thus, low strict-poisoning rates reported for negation families represent a mechanical measurement boundary rather than definitive evidence of model resilience. Finally, because discourse markers are stripped before parsing, we exclude and separately count unparsed runs to account for instances where the model breaks the parser by copying injected phrasing.

Example. Table 2 shows one audited PrOntoQA instance. The relevant chain is *Stella is a lorp* → *Stella is a shumpus* → *Stella is a wumpus* → *Stella is not liquid*; the full prompt also contains distractor facts and a second premise (*Stella is a dumpus*). The same world produces all four validator outcomes depending on the planted step and the continuation.

Condition	Planted step	Continuation excerpt	Validator
1-hop falsehood	Stella is not mean.	“Contradiction, but not needed for proof. Stella is a dumpus. Every dumpus is not opaque. ...Stella is not liquid.”	closure-valid
global falsehood	Stella is an impus.	“Every impus is not red. Stella is not red. ...Stella is a wumpus. Every wumpus is a sterpus. ...”	injection-dependent
benign paraphrase	Thus, Stella is not opaque.	The model eventually states the target, but inserts underivable category steps on the way.	parroted
1-hop falsehood	Stella is not a dumpus.	The model continues with other facts and never ends on <i>Stella is not liquid</i> .	derailed

Table 2: A single instance illustrating the validator categories. For genuinely false, positive-category injections, injection-dependence is strict poisoning.

3 Results

3.1 Nearby Evidence and Recovery

Table 3 presents the primary outcomes regenerated from validated artifacts. Because any interruption can lower closure-valid completion, we interpret all perturbations against the benign-paraphrase baseline. The central contrast is whether the surrounding proof contains nearby evidence against the planted error. Errors contradicted by nearby statements are more often flagged or avoided; errors requiring a broader proof search are more often used in later steps. The table also shows exceptions and parser failures, so the result is evidence for this nearby-evidence pattern rather than proof of a single mechanism.

Family	Inj.	n	Closure-valid [95% CI]	Inj.-dep.	Parroted	Doubt	Unparsed
benign paraphrase	early	150	0.747 [0.672,0.810]	—	0.027	0.087	0.053
benign paraphrase	mid	150	0.880 [0.818,0.923]	—	0.040	0.127	0.020
benign paraphrase	late	150	0.920 [0.865,0.954]	—	0.000	0.027	0.000
true interruption	early	150	0.820 [0.751,0.873]	0.000	0.127	0.007	0.020
true interruption	mid	150	0.867 [0.803,0.912]	0.000	0.100	0.027	0.013
true interruption	late	150	0.880 [0.818,0.923]	0.000	0.100	0.027	0.007
distractor rule	early	130	0.731 [0.649,0.800]	0.000	0.100	0.023	0.054
distractor rule	mid	130	0.715 [0.633,0.786]	0.000	0.138	0.031	0.092
distractor rule	late	130	0.762 [0.681,0.827]	0.000	0.146	0.031	0.054
contradiction (dist. 0)	early	130	0.731 [0.649,0.800]	0.000	0.108	0.231	0.077
contradiction (dist. 0)	mid	130	0.792 [0.715,0.853]	0.000	0.023	0.269	0.100
contradiction (dist. 0)	late	130	0.854 [0.783,0.904]	0.008	0.023	0.177	0.069
false, 1-hop checkable	early	150	0.633 [0.554,0.706]	0.000 [†]	0.147	0.467	0.047
false, 1-hop checkable	mid	150	0.640 [0.561,0.712]	0.020 [†]	0.213	0.273	0.013
false, 1-hop checkable	late	150	0.740 [0.664,0.804]	0.027 [†]	0.087	0.267	0.053
false, globally checkable only	early	150	0.447 [0.369,0.527]	0.233	0.233	0.007	0.027
false, globally checkable only	mid	150	0.360 [0.288,0.439]	0.373	0.193	0.013	0.013
false, globally checkable only	late	150	0.373 [0.300,0.453]	0.420	0.173	0.020	0.013

Table 3: Formally validated outcomes for the main Qwen2.5-7B PrOntoQA sweep. Cell sizes, rates, and confidence intervals are regenerated from validated result artifacts; rows with $n = 150$ use the expanded audited sample, while the distractor and contradiction rows retain the original auxiliary sweep. “Closure-valid” is semantic validity under rule closure, not strict step-by-step replay. “Inj.-dep.” (injection-dependent) indicates that a subsequent reasoning step logically follows only if the planted atom is assumed to be true; for genuinely false positive-category claims, this constitutes strict error poisoning. [†]For negated injections, this validator flag is not strict poisoning.

Errors requiring broader proof search. When a planted falsehood is detectable only by checking the full rule closure, the model often completes a confident-looking proof and uses the planted fact in later derivation steps, with little explicit doubt.

Errors with nearby evidence against them. When a falsehood can be checked against adjacent context, explicit doubt is largest among the main families, while closure-valid completion remains substantially above the global falsehood family. The model’s response

is associated with how easy the error is to check against the surrounding proof, not merely with whether it is false.

Problem-clustered robustness. The same pattern survives a fixed-effect logistic model fit to the raw Qwen2.5-7B validation rows with injection position included and problem IDs resampled as clusters. Relative to benign paraphrase, both one-hop falsehoods and errors requiring broader proof search reduce the adjusted odds of closure-valid completion, with the global falsehood family showing the larger reduction. For explicit doubt, the direction reverses when nearby evidence is available: contradiction and one-hop falsehood increase doubt, while global falsehoods remain below the benign baseline. Intervals are problem-cluster bootstrap intervals stored in the cluster-statistics artifact.

A Dose-Response in Inferential Distance. The availability of nearby evidence can be varied parametrically. By negating a fact the model will derive k hops downstream, we place the contradicting evidence exactly k inferences away. At early injection positions, explicit doubt and closure-valid completion decline as the evidence moves farther away, then plateau for longer chains. At the 32B scale, the curve shifts upward and compresses, indicating that scale lifts absolute robustness without removing the distance dependence.

Verification prompting does not reliably fix errors without nearby evidence. Explicitly instructing the model to verify facts against the rules before using them fails to mitigate the dangerous case in the available verification run. At the middle injection position, globally-checkable falsehoods still show injection-dependence 0.371 and closure-valid completion 0.343 ($n = 35$), with only 0.029 verbalized doubt. By contrast, one-hop falsehoods under the same prompting condition produce 0.423 doubt ($n = 130$). The instruction raises visible caution most clearly when the lie can already be checked against nearby context; it is easier to induce cautionary language than to induce checking of claims with no nearby evidence against them.

3.2 Targeted Artifact-Locked Controls

Following the primary sweep, we executed two strictly paired, artifact-locked control experiments to probe the role of local evidence and rule out potential lexical confounds.

Local-Certificate Flip. We tested whether artificially injecting local evidence changes the model’s response to an otherwise global falsehood. This intervention inserts a local “certificate” that directly contradicts the planted global claim. Across 100 paired triples at each injection position (183 problem clusters), each condition contributes 300 continuations. The presence of the certificate lowered injection-dependence ($\Delta = -0.127$) and strongly increased explicit doubt ($\Delta = +0.370$). However, it did not guarantee full proof repair: closure-valid completion improved only marginally ($\Delta = +0.090$), accompanied by an increase in parser-breaking formatting. The certificate primarily changes error-flagging behavior and reduces injection-dependence; it does not cleanly improve closure-valid completion.

Condition	n	Closure-valid [95% CI]	Inj.-dep. [95% CI]	Doubt [95% CI]	Unparsed
global baseline	300	0.377 [0.317,0.436]	0.153 [0.112,0.201]	0.010 [0.000,0.023]	0.040
local certificate	300	0.467 [0.402,0.538]	0.027 [0.010,0.043]	0.380 [0.319,0.445]	0.317
irrelevant certificate	300	0.500 [0.436,0.564]	0.240 [0.187,0.298]	0.023 [0.010,0.041]	0.037

Table 4: Local-certificate flip. The local certificate makes falsifying evidence immediately available before the same globally-checkable falsehood; the irrelevant certificate is length-matched and true but does not locally falsify the falsehood. Confidence intervals are problem-cluster bootstrap intervals.

Lexical-Polarity Control. To isolate logical locality from mere lexical polarity (affirmative versus negative phrasing), we attempted a 2×2 locality-by-polarity sweep. The rigid

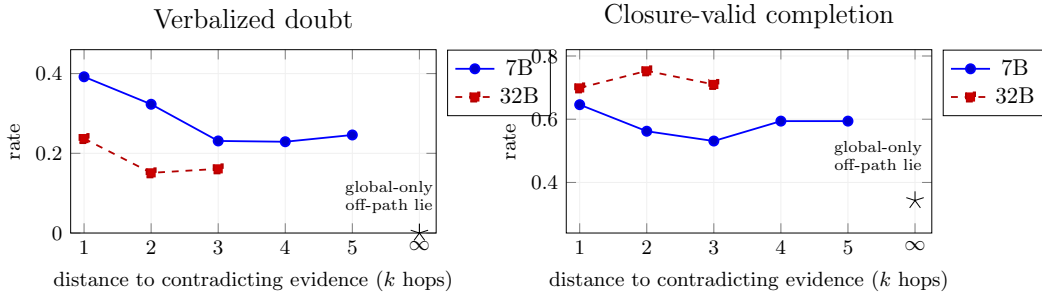


Figure 1: **The response to a planted lie is a dose-response in inferential distance.** Negating the fact the model will derive k hops downstream places the falsifying evidence k inferences from local context. Both verbalized doubt (left) and closure-valid completion (right) decline as k grows and then plateau ($k \geq 3$); globally-checkable-only lies (off-path, $k \rightarrow \infty$, $\star$) are often used without explicit flagging. 32B shifts the curve upward rather than removing the distance pattern.

ontology grammar prevented a clean local, affirmative false positive without changing the category rules, bottlenecking the fully matched cohort to 7 problems (84 validated rows). While underpowered, the available data align explicit doubt with local evidence rather than polarity. The closure-valid signal, however, remains too noisy for formal inference.

Condition	n	Closure-valid [95% CI]	Doubt [95% CI]	Inj.-dep. [95% CI]	Unparsed
local positive	21	0.714 [0.476,0.905]	0.286 [0.095,0.476]	—	0.095
local negative	21	0.524 [0.286,0.762]	0.333 [0.095,0.571]	—	0.048
global positive	21	0.667 [0.429,0.905]	0.000 [0.000,0.000]	0.238 [0.048,0.476]	0.000
global negative	21	0.714 [0.524,0.905]	0.143 [0.000,0.333]	—	0.000

Table 5: Lexical-polarity control. The strict categorical local-positive cell was infeasible in this grammar; the generated local-positive cell uses affirmative attributes, so the table is reported as a boundary-condition check rather than a decisive lexical control. Confidence intervals are problem-cluster bootstrap intervals.

3.3 Strict Stepwise Validation

The primary proof validator is closure-based: it checks whether every generated fact follows from true premises and prior valid steps, but allows skipped intermediate steps. The lexical-polarity control additionally stores a strict replay validator that asks whether the continuation follows the original gold suffix step by step. This stricter audit is reported separately in Table 6; crucially, it never overwrites the closure validator used for our headline metrics.

Control	n	Strict replay?	Closure-valid	Strict final target match	Strict final mismatch
Lexical-polarity control	84	yes	55 (0.655)	66 (0.786)	18 (0.214)
Local-certificate flip	900	no	403 (0.448)	—	—

Table 6: Strict stepwise validation audit for new controls. Counts are shown with rates in parentheses. The strict replay audit records whether the continuation reaches the stored final target under the suffix comparison; it is not substituted for the closure validator used in headline metrics and is reported only where implemented.

3.4 Task Derivation Structure Dictates Recoverability

We applied the same mid-trajectory corruption protocol across three distinct task structures. The pattern tracks the logical redundancy of the task. In PrOntoQA, multiple derivation

197 paths can reach the goal, so a planted error can sometimes be routed around. In chained
 198 arithmetic, every intermediate value is load-bearing and there are no redundant paths, so
 199 an error that the model uses almost always propagates to the next step.

200 This comparison clarifies what “easy to check” means here. In GSM8K, the corrupted value
 201 is technically checkable by recomputation, yet Table 7 shows that it is usually used. The
 202 PrOntoQA result depends on facts explicitly visible in the context window, not on arithmetic
 203 the model must spontaneously redo. Models appear more likely to re-check information
 204 they can re-read than information they must recompute, so recoverability depends on task
 205 structure as much as model architecture.

Task	Closure-valid	Inj.-dep.	n
PrOntoQA (redundant derivations)	0.360	0.373*	150
GSM8K (natural-language math)	0.088	0.877	57
Chained arithmetic (no redundancy)	0.020	0.990	99

Table 7: Mid-trajectory corruption across task structures. Numbers are regenerated from the task summary artifacts; the PrOntoQA row uses the expanded globally-checkable falsehood cell. *In PrOntoQA, absorption is measured as injection-dependence under the proof validator; in arithmetic tasks, absorption corresponds to propagation of the corrupted value.

206 3.5 Goal-Anchor Copying Does Not Fully Explain Recovery

207 To rule out the possibility that apparent recovery is merely the model re-reading and copying
 208 the target statement provided in the prompt, we conducted two targeted ablations. First,
 209 we obscure the target’s truth value by reframing each problem as a boolean query (“True
 210 or false: s ?”, where s is the true target in half the instances and its negation in the other).
 211 This exposes the statement but removes the definitive answer to copy. Under this constraint,
 212 closure-valid completion is attenuated but persists in the available pilot data, showing that
 213 goal-copying heuristics cannot fully explain the error-recovery phenomenon.

214 Second, we applied a direct attention mask to the goal span. This intervention collapsed
 215 closure-valid completion across both the falsehood and benign arms. Because masking the
 216 goal removes the model’s structural stopping condition, this manipulation fundamentally
 217 breaks the task rather than isolating the recovery mechanism, yielding a negative result.
 218 However, it provides one critical qualitative insight: goal access facilitates overall derivation
 219 quality, rather than specifically driving error repair.

220 3.6 Latent Signatures and the Stylistic Confound

221 We investigated whether error recovery leaves a detectable signature in hidden states and
 222 found a strongly negative result. Jensen–Shannon divergence (ρ_h^{func}) between perturbed
 223 and gold next-token distributions spikes immediately after the injection, but fails to track
 224 semantic content. The benign paraphrase — a strictly stylistic edit — produced the largest
 225 early divergence of any perturbation family. Because the model adopts the injected phrasing
 226 for the rest of its proof, the metric registers surface style as heavily as a planted logical lie.

227 Subsequent controls confirmed this is a geometric confound rather than a sampling arti-
 228 fact. Final-layer cosine divergence remained statistically indistinguishable between benign
 229 paraphrases and verified falsehoods. As instrumented, trace-based divergence cannot dis-
 230 tinguish a logical error from a stylistic change. Our primary claims therefore remain strictly
 231 behavioral and formally validated.

232 3.7 Scale and Lineage: Explicit Doubt and Recovery Dissociate

233 Scaling Qwen2.5 from 1.5B to 32B suggests that scale improves recovery from errors with
 234 no nearby evidence against them and reduces how often the continuation uses the planted
 235 error, but it does not make explicit doubt a reliable proxy for closure-valid completion.

Model	globally-checkable falsehood			1-hop falsehood
	closure-valid	inj.-dep.	doubt	doubt
Qwen2.5-1.5B	0.154–0.462	0.308–0.615	0.000–0.000	0.000–0.018
Qwen2.5-7B (primary)	0.360–0.447	0.233–0.420	0.007–0.020	0.267–0.467
Qwen2.5-32B	0.600–0.733	0.100–0.367	0.033–0.067	0.129–0.280

Table 8: Scale sweep over Qwen2.5 models. Ranges are over early, middle, and late injection positions and are regenerated from summary artifacts. The Qwen2.5-7B row uses the expanded primary artifact, making it numerically comparable to Table 3; the other size rows use the matched cross-model summary artifacts.

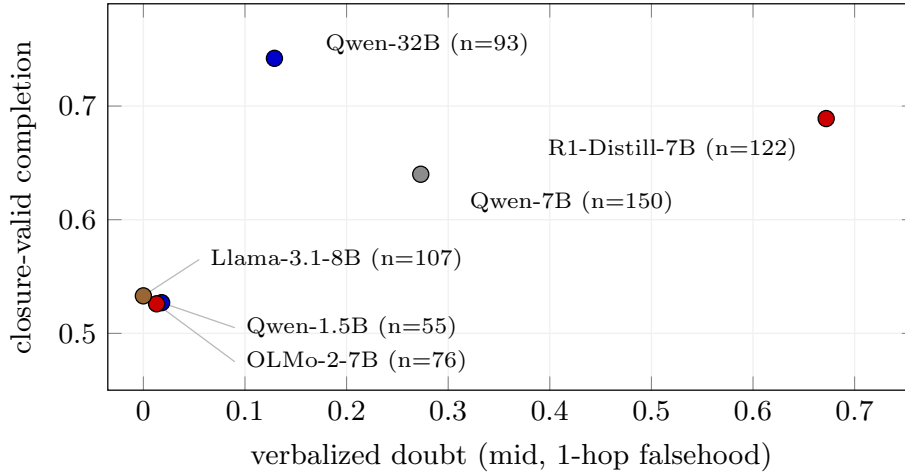


Figure 2: **Explicit doubt and recovery dissociate.** Across six models (1-hop falsehood, mid injection), verbalized doubt varies much more than closure-valid completion. Points are cell rates with the per-model n printed on the labels. The Qwen-7B point uses the expanded primary artifact; the other points use the matched cross-model summaries. Reasoning-RL training (R1-Distill) increases explicit flagging at flat closure-valid completion; the base/instruct models recover at comparable rates while saying little. Verbalizing a caught error and correcting it are dissociable behaviors.

236 The same nearby-evidence pattern appears across architectural lineages. Both OLMo-2-
237 7B-Instruct and Llama-3.1-8B-Instruct show the behavioral gradient in the cross-model
238 summary artifacts. Crucially, both models can recover from errors with nearby evidence
239 against them while showing little explicit doubt, indicating that successful error repair does
240 not require the model to explicitly flag the mistake.

241 Finally, evaluating DeepSeek-R1-Distill-Qwen-7B isolates the impact of reasoning-specific
242 reinforcement learning. While closure-valid completion remains broadly in the same range
243 as its base model, explicit flagging of one-hop falsehoods is much higher. Across the model
244 set, explicit flagging changes more than closure-valid completion. We therefore treat explicit
245 doubt as a trainable interface behavior and an unreliable proxy for closure-valid completion.

246 4 Limitations

247 Our coverage is still narrow. The tasks are two synthetic families plus GSM8K; a formal-
248 verification task such as Lean is the natural next regime. The models span several Qwen
249 sizes, two further families (OLMo-2 and Llama-3.1), and a reasoning-RL variant (R1-Distill),
250 but the model set is still small and Qwen-heavy. We also use deterministic decoding and a
251 single null-sampling seed.

Several design limitations remain. The globally-checkable falsehood family is generation-constrained, although the primary sweep expands the matched validated sample to 150 examples per injection position. The lexical-polarity control does not cleanly isolate polarity because the local false positive categorical cell is infeasible in the current grammar. The prompt states the proof target, which the validator’s parroting category and the goal-hidden QA condition address only partially. Our latent and geometric metrics are negative results. Closure-based validation permits step-skipping; the strict replay audit is reported separately and should not be conflated with closure-valid completion.

5 Related work

CoT interventions and error recovery. von Recum et al. (2026) perturb visible CoT in reasoning-tuned models and find behavioral robustness with early-intervention degradation and doubt/length effects; our truth-status audit suggests that “adversarial” interventions need per-injection falsity verification. Yee et al. (2024) precede our closure-valid/parroting split with a faithful/unfaithful recovery dichotomy via manual annotation; our delta is automated formal validation, semantic (vs. numerical) error control, the truth-status audit, and the comparison between errors with nearby evidence and errors requiring broader proof search. Their finding that recovery is more frequent for *obvious* errors independently anticipates this distinction.

Lanham et al. (2023) introduced adding-mistakes interventions for faithfulness; Turpin et al. (2023) showed stated reasoning can omit operative causes, which is consistent with models continuing without flagging an error they have used. Zhang et al. (2023) showed models commit to early mistakes they can separately recognize; our results locate when that recognition is and is not deployed in-rollout. Huang et al. (2024) found intrinsic self-correction weak at the multi-turn level. Our single-pass setting shows both sides: models may fail to flag errors without nearby evidence, but can explicitly flag inconsistencies when the relevant evidence is local.

Self-repair and redundancy. McGrath et al. (2023); Lad et al. (2024) document structural self-repair. Our results describe the complementary failure mode: robustness machinery can smooth over content errors that are not easy to check against the surrounding proof.

Trajectory geometry. Sun et al. (2026) characterize reasoning as step-specific representational trajectories with correctness signals. Our style-propagation result (benign $\approx$ falsehood in both functional and geometric divergence) is a caution for cross-trace comparisons in that line.

Compounding. Dziri et al. (2023); LeCun (2022) supply the ingredients of the compounding frame our results sharpen: the model can build on a planted error when the surrounding proof lacks nearby evidence against it, and task derivation structure (§3.4) determines the price.

6 Conclusion

Single-pass autoregressive reasoning is neither uniformly brittle nor reliably self-correcting. In our controlled setting, recovery depends partly on whether the planted error can be checked against nearby context. Errors contradicted by nearby statements are often survived; errors that require a broader proof search are more often used without being flagged. This reconciles part of the compounding-error argument with robustness results: compounding is a better failure model when the surrounding proof lacks nearby evidence against the error, but it overstates fragility when the contradiction is easy to recover from nearby context. Explicitly flagging an error and actually recovering from it are different behaviors. For evaluation, this means that planted-error studies should specify not only whether an injected step is false, but how locally that falsehood can be discovered. For model design, counterevidence should be salient before a bad intermediate state is reused.

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